Intelligence Academy

Scikit-Learn 101

09. Fit for supervised models With Python

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I. The Supervised Learning Paradigm

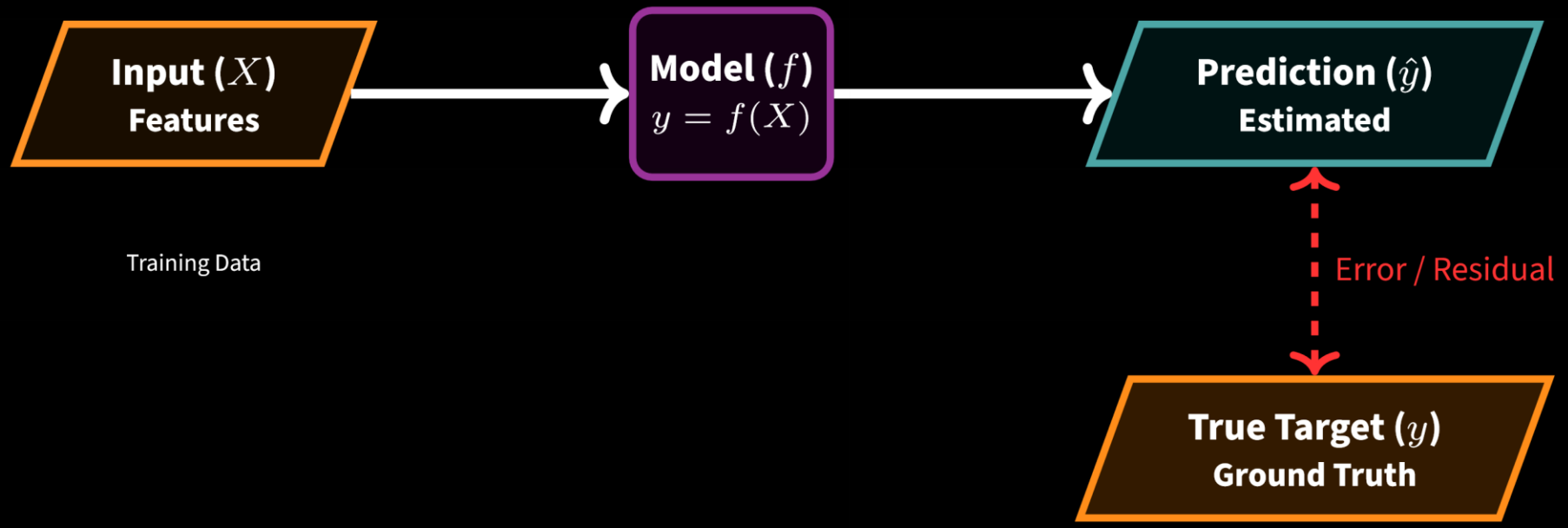
Goal: Mapping Inputs (X) to Targets (y)

Core Concept: Supervised learning is the process of learning a function that maps an input to an output based on example input-output pairs.

$$\hat{y} = f(X; \theta) \approx y$$

- **Features (X):** The input variables (matrix of shape $N \times D$).
- **Targets (y):** The "ground truth" labels we want to predict.

The objective is to find the parameters θ such that the predictions $\hat{y}$ are as close as possible to the true targets y for unseen data.



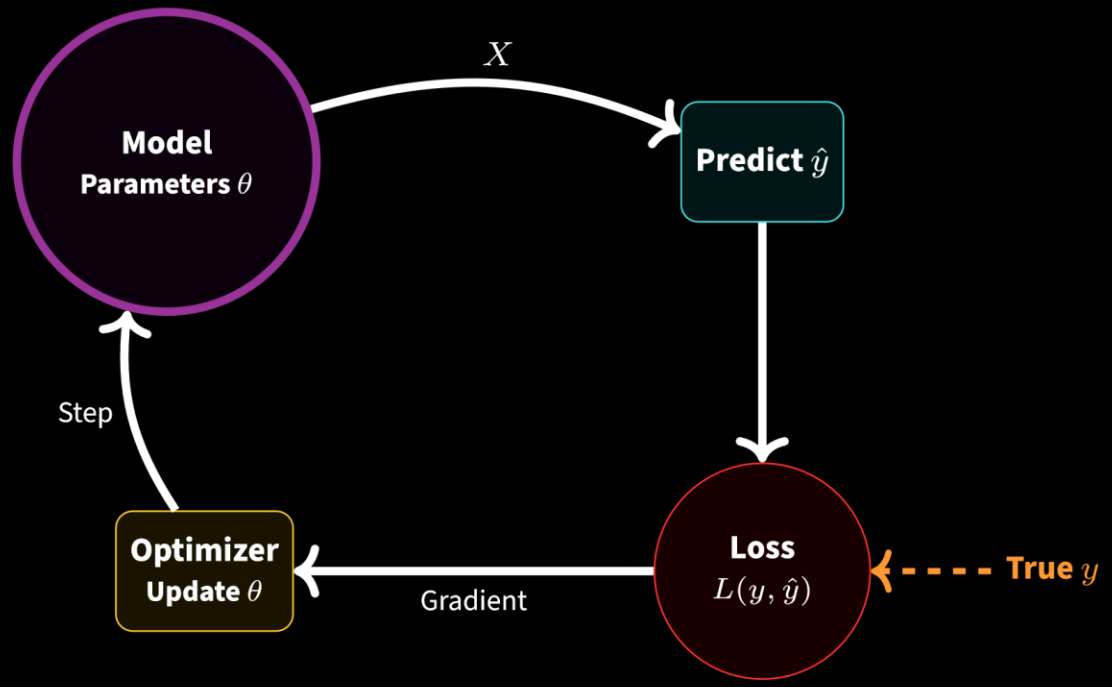
The Role of the Loss Function in Training

How do we measure "success"?

- **The Metric:** A **Loss Function** $L(y, \hat{y})$ quantifies the penalty for bad predictions.
- **The Goal:** Training is an optimization problem where we minimize this loss.

$$\min_{\theta} \sum_{i=1}^{N} L(y^{(i)}, f(X^{(i)}; \theta))$$

The `.fit()` method runs an iterative loop: **Predict** $\rightarrow$ **Calculate Loss** $\rightarrow$ **Update Model**.



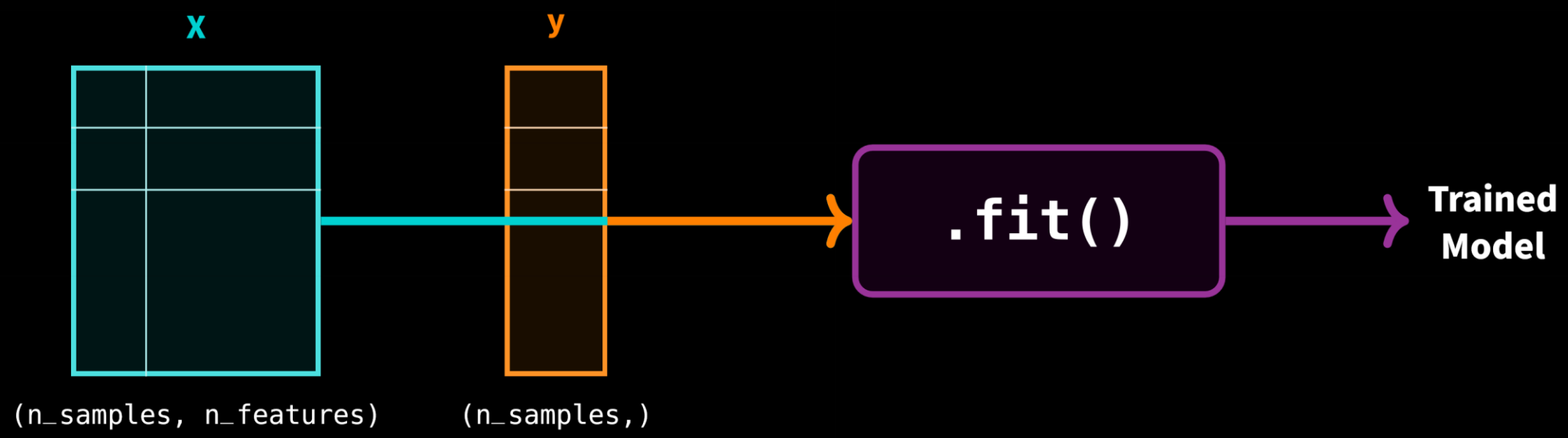
II. The `.fit(X, y)` Method

Standard Syntax and Arguments

In Scikit-Learn, the API is consistent across all supervised models. The training process is always triggered by the `fit` method.

```
model.fit(X, y)
```

- **X (Features):** A 2D array-like structure (Matrix).
  Shape: `(n_samples, n_features)`
- **y (Target):** A 1D array-like structure (Vector).
  Shape: `(n_samples,)`
- **Returns:** The estimator object itself (allows method chaining).



Statefulness: How a Model Changes

Before vs. After: A Scikit-Learn model is "stateful." Calling `.fit()` mutates the object by adding new attributes.

1. **Initialization (Constructor):**
   Only stores Hyperparameters (configuration).
   Example: `LinearRegression(fit_intercept=True)`
2. **After Fitting:**
   Calculates and stores Learned Parameters.
   Naming Convention: Learned attributes always end with an **underscore** (_).
   Example: `model.coef_`, `model.intercept_`

```
// State 1: Unfitted
LinearRegressionObject
------------
fit_intercept = True
n_jobs = None


coef_ <Not Exists>
intercept_ <Not Exists>
```


```
// State 2: Fitted
LinearRegressionObject
------------
fit_intercept = True
n_jobs = None


coef_ = [0.45, -1.2]
intercept_ = 3.14
```

IV. Python Implementation

Step-by-Step Code Walkthrough

The Universal Pattern: Almost every Scikit-Learn model follows this exact 3-step recipe. Memorizing this unlocks 90% of the library.

1. **Import:** Bring in the specific class you need.
2. **Instantiate:** Create an object and set Hyperparameters.
3. **Fit:** Feed the data (X, y) to learn Parameters.

```
# 1. Import the class
from sklearn.linear_model import LinearRegression

# 2. Instantiate the model (Hyperparameters)
model = LinearRegression(fit_intercept=True)

# 3. Fit the model to data (Learning)
# X_train must be 2D, y_train 1D
model.fit(X_train, y_train)
```

The model object is now trained and holds the state.

Case Study: Training LinearRegression

Let's train a model on simple dummy data to verify the shapes and attributes.
Goal: Learn the relationship $y = 2x$.

- X: `[[1], [2], [3]]` (Shape 3×1)
- y: `[2, 4, 6]` (Shape 3)

```python
import numpy as np
from sklearn.linear_model import LinearRegression

# Create dummy data (Note X is 2D)
X = np.array([[1], [2], [3]])
y = np.array([2, 4, 6])

# Train
regr = LinearRegression()
regr.fit(X, y)

# Check learned attributes (suffixed with _)
print(f"Slope: {regr.coef_}")       # Output: [2.]
print(f"Bias:  {regr.intercept_}")  # Output: 0.0
```

✓Training Successful
coef_ learned accurately

Logic:
Input $x = 1 \rightarrow 2(1) + 0 = 2$
Input $x = 2 \rightarrow 2(2) + 0 = 4$

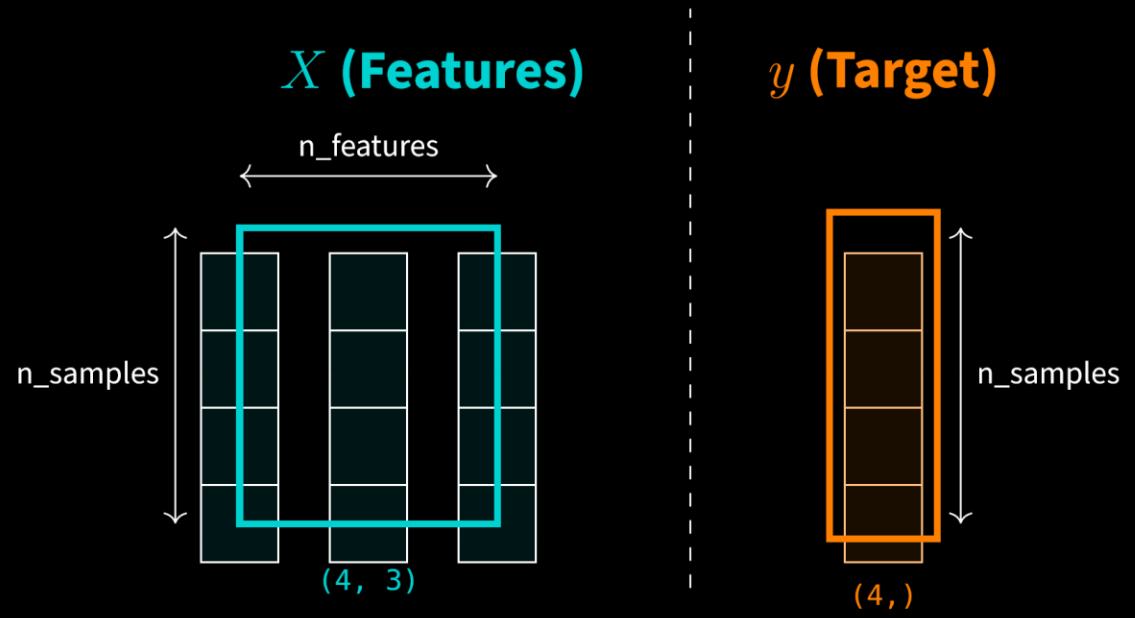
V. Data Shape & Dimensionality

Critical Constraints: The Shape Rules

Scikit-Learn is strict about data dimensions. The `.fit()` method assumes specific mathematical structures for efficient linear algebra operations.

- **Feature Matrix (X): MUST be 2D.**
  Shape: `(n_samples, n_features)`
  Even if you only have **one feature**, it must be a column vector, not a flat list.

- **Target Vector (y): Usually 1D.**
  Shape: `(n_samples,)`
  A simple flat array of labels or values.



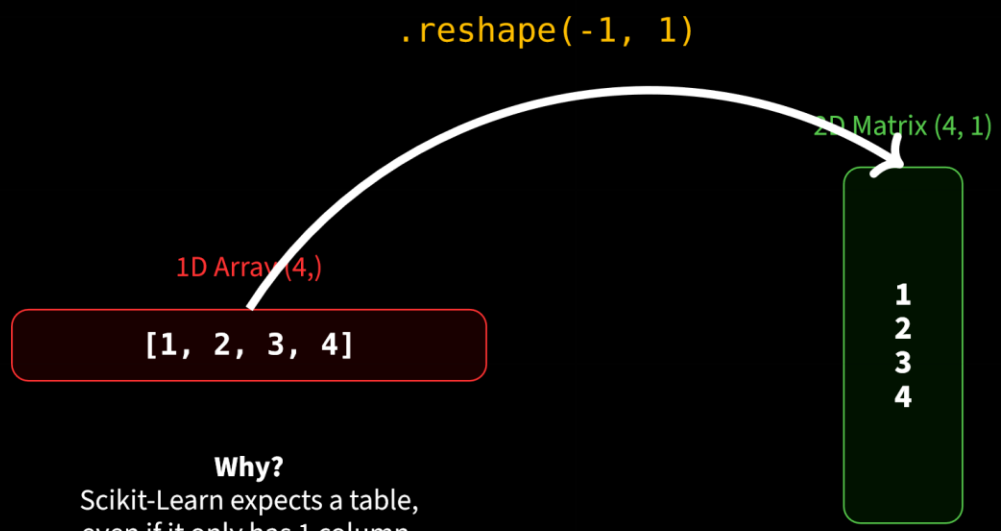
Common Error Handling: Reshaping

The Error: If you pass a 1D array (e.g., `[1, 2, 3]`) as X, you will see: `ValueError: Expected 2D array, got 1D array instead.`

The Fix: Reshape your data to add a second dimension.

```
import numpy as np
x_bad = np.array([1, 2, 3, 4])  # Shape (4,) -> Error!

# Reshape to (n_samples, 1)
# -1 means "calculate this dimension automatically"
X_good = x_bad.reshape(-1, 1)   # Shape (4, 1) -> OK!
```



Why?
Scikit-Learn expects a table,
even if it only has 1 column.